

When Does Fourier Diagonalize the Physics?

Calibrated Meters for Fixed versus Learned Representations in Classification and in Operator Learning, and What the Learned Part Is Worth

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Abstract

A previous study built a classifier from a pooled dictionary of fixed transforms and a learned unitary basis, coupled by a gate that decides how much of the learned branch to use. This paper asks what that gate is *worth*, in two settings that share the architecture but invert its objective. **Classification.** On a family of tasks whose class-separating basis rotates continuously from Fourier to a random orthogonal basis, the gate on the learned branch follows the information the learned branch adds about the label, $G = I(T_{\text{learned}}; Y) - I(T_{\text{dict}}; Y)$ in bits, with a rectified law: shut ($g \leq 0.002$) wherever $G \leq 0$, then rising and saturating as $g \approx 0.59 (1 - e^{-[G]_+/0.30})$ (Pearson $r = 0.99$ on the family, 0.98 on the four held-out benchmark tasks of the previous study). Selecting the same $K = 16$ dictionary coefficients by energy, as a codec would, instead of by task relevance costs 0.265 versus 0.999 accuracy on a localized task: compressing toward the signal and toward the label are different objectives. And the learned classification basis cannot be read back: a closed-form fit that scores the untrained DFT at 1.000 scores the task-trained basis at 0.030. **Physics.** Re-instantiating the architecture for one-dimensional operator learning, where the objective is to predict the state, makes the learned part legible. (i) The failure of the fixed Fourier basis to diagonalize a diffusion propagator is a monotone heterogeneity meter (off-diagonal fraction $0 \rightarrow 0.178$) while the true eigenbasis stays diagonal. (ii) That eigenbasis is discovered without an oracle by minimizing off-diagonality over a Sturm–Liouville family, reaching 0.000000 at every heterogeneity and returning the physical coefficient at correlation 1.0000; the same objective flags its own mis-specification. (iii) The residual of the best data-driven linear propagator — dynamic mode decomposition (DMD), which we show is a data-driven Laplace transform — is a nonlinearity meter, and a gated learned branch fires with the *same rectified, thresholded law* as the classification gate. (iv) Depth is not the lever: a black-box branch captures 45.1% of what the linear propagator misses and deep residual integrators 68.8%, while a structure-matched quadratic branch with no hidden layer captures 90.3%, and the same term sub-stepped four times 98.9%. (v) Sparse regression names that term exactly ($\nu = 0.05$ recovered as 0.050; nonlinear coefficient on the identity line). (vi) The named term buys extrapolation, but only when it is integrated rather than applied in one explicit step: over a 30-step rollout never seen in training the structured integrator is $19\times$ more accurate than the linear model and $31\times$ more than the black box, which diverges; at $1.6\times$ the training amplitude it is $9\times$ and $30\times$ better, and it never creates energy, while the one-step structured surrogate diverges at any amplitude above training. Rollout training then improves it a further $1.6\times$. Two settings are two, not a universal law; what they share is the functional form of the gate and the controls that keep it honest.

1 Introduction

Two families of front-end compete whenever a signal must be represented compactly. A *fixed* transform — Fourier, wavelet, scattering — is cheap, interpretable and needs no data. A *learned*

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transform is adaptable but under-determined and opaque. A companion study [10] compared them for classification and found that a gated coupling of the two behaves best, with the gate acting as a diagnostic: it collapses to the dictionary when the dictionary suffices and engages the learned branch only where the discriminative structure is genuinely non-classical.

That study established *that* the gate works. It left two questions open. First, what does the gate *measure*? A number between 0 and 1 that "opens where needed" is a description, not a calibration. Second, what does the learned branch *say*? Under a classification objective many bases separate the same labels, so the learned basis drifts and its entries carry no stable meaning. This paper answers both, and the answer to the second is what motivates the paper's structure: reading the learned part becomes possible when the objective is inverted from compressing toward a label to predicting a physical state.

One gate, two settings. The gate is the same object throughout: a scalar multiplying a branch, initialized open and pushed shut by an L_1 penalty, so that the branch stays on only if it earns its keep. In classification we show it pays for *information*: it follows the gap in bits between what the learned and the fixed features tell about the label, with a rectified linear law. In physics we show it pays for *nonlinearity*: it stays shut while a linear (Laplace/Koopman) propagator suffices and opens in proportion once the dynamics leave that regime. The two laws have the same shape — a threshold, then proportionality — and we are careful about what that does and does not establish (Section 16).

Contributions. (1) A calibrated reading of the classification gate as an information meter, on a family with a gap controlled by design and on the benchmark tasks of [10]. (2) Two negative results on the classification side that motivate the inversion: energy-based (codec-style) selection discards the discriminative coefficients, and the task-trained basis is not symbolically readable. (3) A set of physical meters with the same shape — fixed structure plus a measured failure — for heterogeneity and nonlinearity, together with an oracle-free basis discovery that returns the physical coefficient and announces its own mis-specification. (4) Evidence that the learned branch is worth most when it is structurally right rather than deep, that the right structure is exactly the term sparse regression names, and that the named term extrapolates in horizon, in amplitude and under rollout training — provided it is integrated, not applied in one explicit step; depth in the service of structure is what buys the stability margin. (5) A list of the artefacts, unfair comparisons and diverging recipes we had to remove, and the controls that replaced them.

2 The architecture and its two gates

Figure 1 shows the pipeline shared by both settings, and Figures 2 and 3 zoom into the two stages that carry a gate. Stage 1 represents the input in two ways at once: a fixed branch (a pooled dictionary of classical transforms in classification; the Fourier basis in physics) and a learned branch (a unitary $U = \exp(C - C^H)$ warm-started at the DFT in classification; a searched Sturm–Liouville basis in physics). Stage 2 keeps K coordinates. Stage 3 predicts: a linear map plus a gated nonlinear branch. In physics that prediction is a *propagator*: the map that advances the state of the system by one time step, $u(t) \mapsto u(t + \Delta)$ — for a linear PDE $u_t = Lu$ it is the matrix exponential $P = \exp(L\Delta)$, for a nonlinear one it is whatever the numerical integrator does over Δ . There are therefore *two* gates, and it matters which one is read where. The gate g_{s2} of Stage 1 multiplies the learned representation and answers "was a learned basis needed?"; the gate g_{nl} (called g in the physics sections) of Stage 3 multiplies the nonlinear branch of the predictor and answers "was a nonlinear model needed?". Table 1 pins the correspondence.

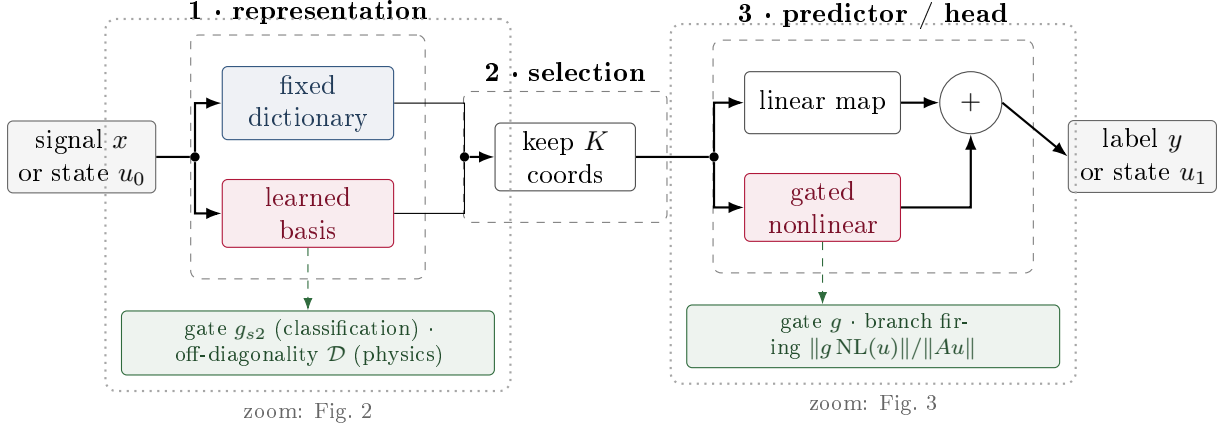


Figure 1: The shared pipeline. Blue: fixed, data-free; red: learned, gated; green: the meters read off the gates. The dotted boxes are expanded in Figures 2 and 3.

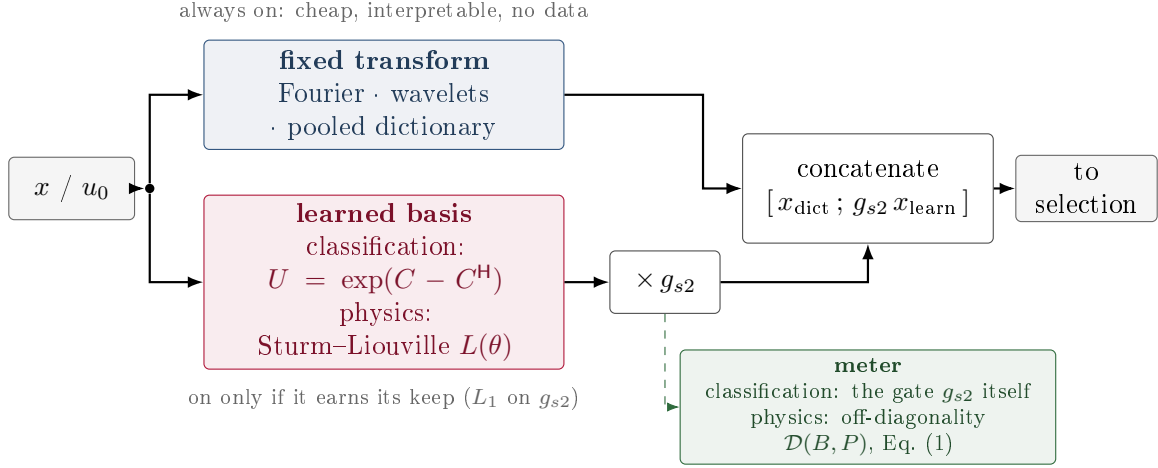


Figure 2: Zoom on Stage 1, the representation. The learned branch enters the concatenation through the gate g_{s2} ; in physics the analogue of U is the searched basis, and the analogue of the gate is the off-diagonality $\mathcal{D}$ of Equation (1).

3 The meters, defined

Frobenius norm. For a matrix M , $\|M\|_F = (\sum_{ij} M_{ij}^2)^{1/2}$ — the Euclidean length of the matrix read as a vector. It is invariant under orthonormal changes of basis, which is what makes the next definition basis-comparable.

Diagonality of a propagator in a basis. For a basis B with orthonormal rows (each row is one basis vector, e.g. one sinusoid of the Fourier basis) and a one-step propagator P (the matrix that advances the state by Δ),

$$\mathcal{D}(B, P) = \frac{\|BPB^\top - \text{diag}(BPB^\top)\|_F}{\|BPB^\top\|_F}, \quad \|BPB^\top\|_F = \|P\|_F. \quad (1)$$

We refer to this as the *off-diagonality* $\mathcal{D}$ throughout. $\mathcal{D} = 0$ means that a per-mode (diagonal) propagator in B reproduces the dynamics exactly: a spectral method is exact and there is nothing to learn. The denominator does not depend on B , so $\mathcal{D}$ compares bases fairly. This is a property of the *operator*, not of sampled states — a distinction that cost us an artefact (Section 16).

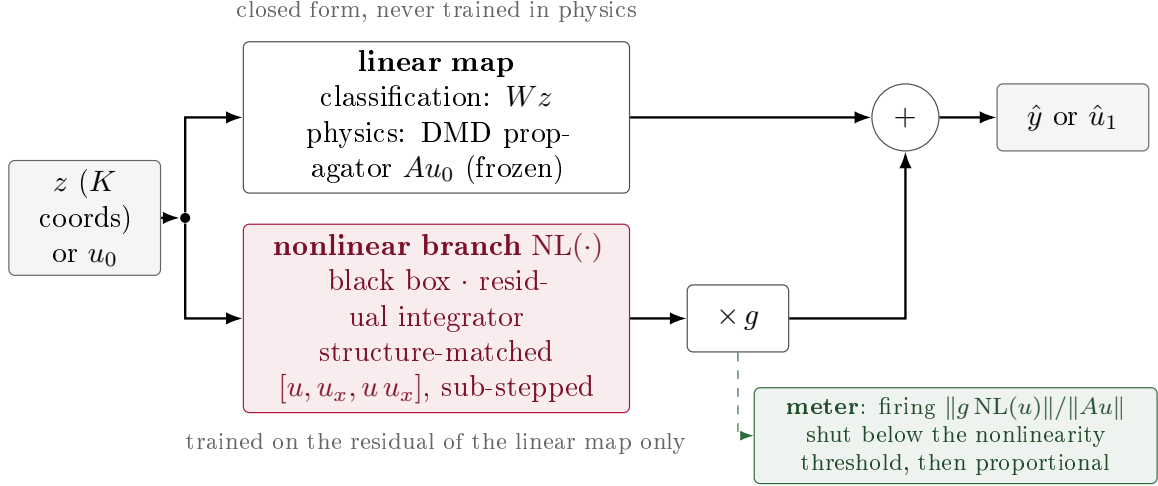


Figure 3: Zoom on Stage 3, the predictor. In physics the linear map is the closed-form propagator fitted by dynamic mode decomposition (DMD, Section 3), frozen; only the gated branch trains, on the residual.

	classification ([10], Part I)	physical modelling (Part II)
objective	compress toward the label	predict / reconstruct the state
fixed branch	Fourier, wavelets, scattering as features	spectral method: Fourier diagonalizes homogeneous operators
learned branch	discriminative unitary U (under-determined)	eigenbasis of the dynamics (determined by the physics)
head	classifier	propagator $u(t) \rightarrow u(t + \Delta)$
gate g_{s2} (Stage 1)	is the discriminative basis outside the bank?	is the physics diagonalized by a classical transform?
gate g_{nl} (Stage 3)	is the head nonlinear?	is the dynamics nonlinear?
reading the learned part	ill-posed: the basis drifts (Section 6)	well-posed: recover the operator (Sections 9, 12)

Table 1: The inversion, and the correspondence between the two settings. Every row changes meaning when the objective changes from label compression to state prediction.

Information gap. For a representation T of the input and a label Y , $I(T; Y) = H(Y) - H(Y|T)$. We estimate $H(Y|T)$ by the held-out cross-entropy of a small head trained on T , which gives a lower bound on $I(T; Y)$ in bits after dividing by $\ln 2$; the bound is clipped to $[0, H(Y)]$. The gap the learned branch adds is $G = I(T_{\text{learned}}; Y) - I(T_{\text{dict}}; Y)$ at equal rate $K = 16$.

DMD as a data-driven Laplace transform. Dynamic mode decomposition [3, 2] fits the best linear one-step map A on snapshot pairs (u_k, u_{k+1}) by least squares. Its eigen-decomposition $A = \sum_j \mu_j \phi_j \psi_j^T$ predicts $u_k \approx \sum_j c_j \mu_j^k \phi_j$: a superposition of exponential modes with continuous-time exponents $\lambda_j = \log \mu_j / \Delta$. Those exponents are the poles of the Laplace transform of the dynamics — a purely oscillatory pole is a Fourier mode ($\sigma = 0$ axis), a real pole is pure decay, a complex pole is a damped oscillation. DMD is therefore the data-driven form of what Prony’s method did for sums of exponentials. It is also the finite-dimensional face of *Koopman theory*: any dynamical system, however nonlinear, acts *linearly* on the space of functions of its state (the Koopman operator advances an observable f to $f \circ \Phi_\Delta$, where Φ_Δ is the flow), so a nonlinear system can in principle be written as a linear one in the right — usually infinite — set of coordinates. DMD is the least-squares approximation of that operator restricted to the coordinates one actually measures; when those coordinates are the state itself, its residual measures how far the dynamics are from linear in them. We checked this literally: on the

heat equation the DMD exponents equal $-\nu m^2$ for every excited mode $m \leq 4$ to $4 \cdot 10^{-10}$ (`dmd_laplace_check.py`). The consequence used throughout Part II is that the held-out residual of A is exactly what a Laplace/Koopman-linear front-end cannot capture, i.e. a nonlinearity meter.

Gate firing. For the predictor $\hat{u}_1 = Au_0 + g \text{NL}(u_0)$ of Figure 3 we report the firing amplitude $\|g \text{NL}(u)\| / \|Au\|$ on held-out data rather than g alone, because a branch can rescale itself against g ; the product is what the model actually uses.

Part I — Classification: what the gate pays for

4 The gate law in bits

A family with a gap controlled by design. Take the $N = 128$ real Fourier basis F and a fixed random orthogonal basis Q , and move along the geodesic $Q(\theta) = F \exp(\theta \log(F^\top Q))$: $\theta = 0$ is Fourier, $\theta = 1$ is a basis no classical transform knows. Four classes are separated by three active coordinates each in $Q(\theta)$, with noise, exactly as in the "hidden basis" task of [10]. At small θ the pooled dictionary sees the classes and the learned branch has nothing to add; as θ grows the dictionary's $K = 16$ selected coefficients lose the classes while a learned unitary, warm-started at the DFT and task-trained, keeps them. For every θ and seed we measure G in bits and the gate g_{s2} of the S3 head of [10], unmodified (L_1 weight 0.25). For readers who have not seen that study: S1 is the pooled dictionary alone (Fourier, wavelet, scattering and related coefficients of the input, from which the $K = 16$ with the highest class F-score are kept), S2 is the learned unitary alone, and S3 is the coupling of Figure 2 — a small head over $[x_{\text{dict}}; g_{s2} x_{\text{learn}}]$ whose gate g_{s2} starts at 1 and is pushed toward 0 by the L_1 penalty. The F-score of a coefficient is the one-way ANOVA ratio of its between-class to within-class variance: a large value means the coefficient separates the classes on its own, so ranking by F-score keeps the most discriminative coordinates rather than the most energetic ones.

Result. Figure 4 shows the law. In all 18 runs with $G \leq 0$ the gate is shut ($g_{s2} \leq 0.002$); where $G > 0$ it opens, steeply at first and then saturating. Fitted on the 36 runs of the family only,

$$g_{s2} \approx 0.59 (1 - e^{-[G]_+/0.30}), \quad r = 0.99 \text{ (family)}, \quad r = 0.98 \text{ (held-out tasks)}, \quad (2)$$

with G in bits. The simpler rectified line $g_{s2} \approx 0.08 + 0.35 [G]_+$ scores $r = 0.91$ on the family and 0.96 on the held-out tasks; the saturation is real and expected — the gate cannot usefully exceed the weight the head assigns to the learned features, and the L_1 penalty caps it. The 12 runs on the four benchmark tasks of [10] were not used in the fit and fall on the curve: the two classical tasks at $G \approx 0$ with the gate shut, the hidden-basis task at $G = 1.34$ bits with $g_{s2} = 0.55$, XOR-in-hidden-basis at $G = 0.31$ bits with $g_{s2} = 0.26$. The reading is the one promised in the title of [10]: the gate does not open by disposition, it pays for the learned branch what that branch adds about the label, and nothing before that.

5 Compressing toward the signal is not compressing toward the label

A codec keeps the coefficients with the most energy; a classifier needs the coefficients with the most class information. With the same pooled dictionary and the same head, selecting $K = 16$ coefficients by energy against by class F-score gives Table 2 (three seeds). On the global (Fourier)

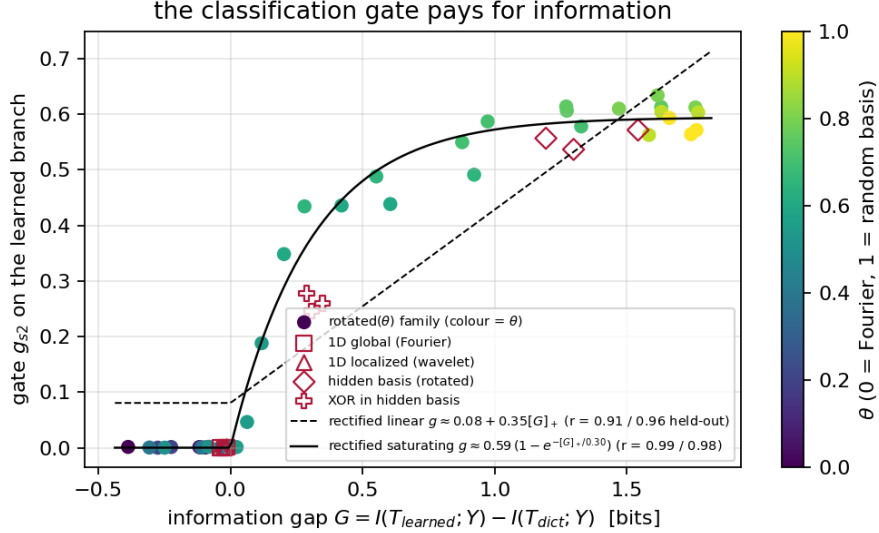


Figure 4: The classification gate against the information gap. Filled circles: the rotated family, coloured by θ ; open markers: the four tasks of [10], not used in the fit. Solid: the gate law, Equation (2); dashed: the rectified line.

task	$K = 16$ by energy	$K = 16$ by class F-score
1-D global (Fourier)	1.000	1.000
1-D localized (wavelet)	0.265	0.999
hidden basis (rotated)	0.584	0.660

Table 2: Test accuracy, mean over three seeds, same dictionary and head. What you keep decides.

task the two selections agree, because the discriminative tones are also the energetic ones. On the localized task energy selection keeps the carrier’s spectral lines and discards the wavelet coefficients that encode *position*, which is the class: accuracy drops from 0.999 to 0.265. On the hidden-basis task both selections fail, because no dictionary coefficient carries the classes — which is exactly where Section 4 showed the gate opening. Rate–distortion and rate–relevance [9] are different objectives, and the gate measures the second, not the first.

6 The classification basis cannot be read back

If a learned unitary is worth something, one would like to *name* it. We tried the simplest reading. A learned unitary is only defined up to a *gauge*: permuting its rows, or multiplying any row by a unit-modulus phase, changes the matrix but not the transform it computes (the feature is $|Ux|$), so two identical transforms can look completely different as matrices. *Gauge canonicalisation* removes that freedom before comparing: rows are sorted by their dominant frequency and each row’s phase is fixed so that its dominant Fourier coefficient is real and positive. After canonicalising, fit the closed form $U_{jk} = e^{-iajk}/\sqrt{N}$ and report the normalized inner product with the best a . The untrained DFT scores 1.000 at $a = 1.000 \cdot 2\pi/N$ — the reading works. A unitary trained with a compression prior plus cross-entropy scores 0.088; one trained by cross-entropy alone from a random start scores 0.030, with 0.0% of its rows still pure sinusoids (Figure 5). The task-trained basis is not a deformed Fourier transform that a formula could name: it is whatever separated the labels, and many bases do. This is the last row of Table 1, and the reason Part II exists. The same operation — read the learned part — becomes well-posed when the learned part is constrained by an operator rather than by a label.

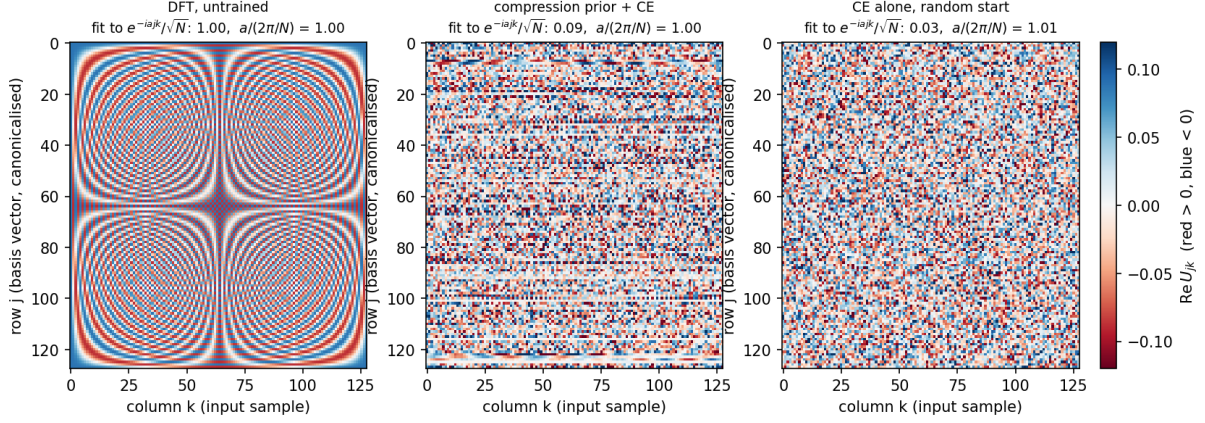


Figure 5: The real part of each matrix entry $\text{Re} U_{jk}$ for three unitaries after gauge canonicalisation (row j , column k ; red positive, blue negative, white zero — the DFT shows its characteristic hyperbolic fringes), with the closed-form reading score. Left: the DFT reads exactly. Centre and right: task-trained bases do not.

Part II — Physics: what the learned part says

7 Setup, and why each choice

Two families of operator. We use two controlled one-dimensional families on a periodic grid of $N = 64$ points on $[0, 2\pi)$. The *linear* family is diffusion with a spatially varying coefficient, $u_t = \partial_x(a(x) \partial_x u)$ with $a = 1 + \beta f(x)$, f a fixed smooth normalized profile and β the heterogeneity knob; $\beta = 0$ is the constant-coefficient heat equation, whose eigenbasis is exactly Fourier. The *nonlinear* family is viscous Burgers, $u_t = \nu u_{xx} - \varepsilon u u_x$ with $\nu = 0.05$ and ε the nonlinearity knob; $\varepsilon = 0$ is the heat equation, $\varepsilon > 0$ steepens fronts into shocks. Each family has a single knob that moves the physics away from the regime where a fixed spectral basis is exact, which is what a meter needs to be calibrated against.

Why divergence form and why $L = -G^\top \text{diag}(a_f)G$. Writing diffusion as $\partial_x(a \partial_x u)$ rather than $a u_{xx}$ keeps the operator self-adjoint, so it has a real spectrum and an orthonormal eigenbasis — the object the fixed and learned bases are compared against. The discretization mirrors that structure: G is the periodic forward-difference matrix, a_f is the coefficient on cell faces, and $L = -G^\top \text{diag}(a_f)G$ is symmetric negative semi-definite by construction. It is also the discrete form of a *Sturm–Liouville operator*, which is why the search family of Section 9 can be written in the same form. Briefly: a Sturm–Liouville problem is the eigenvalue problem $-(p(x)y')' + q(x)y = \lambda w(x)y$ on an interval with boundary conditions — the general self-adjoint second-order differential operator. Its eigenfunctions are real, orthogonal (with weight w) and complete, so they form a basis exactly as the sinusoids do; the sinusoids are the special case $p = w = 1$, $q = 0$ on a periodic interval, and Bessel, Legendre, Hermite and Chebyshev functions are other choices of (p, q, w) . Our diffusion operator is the case $p = a(x)$, $q = 0$, $w = 1$: choosing the coefficient $a(x)$ is choosing which orthogonal basis the physics prefers.

Why a propagator. A learned model predicts states, not operators, so the object to compare is the one-step map $u(t) \mapsto u(t + \Delta)$. For the linear family it is available in closed form, $P = \exp(L\Delta)$, and Equation (1) (the off-diagonality $\mathcal{D}$) can be evaluated on it exactly. For the nonlinear family the propagator is the RK4 integration over Δ , and the best linear approximation to it is what DMD fits.

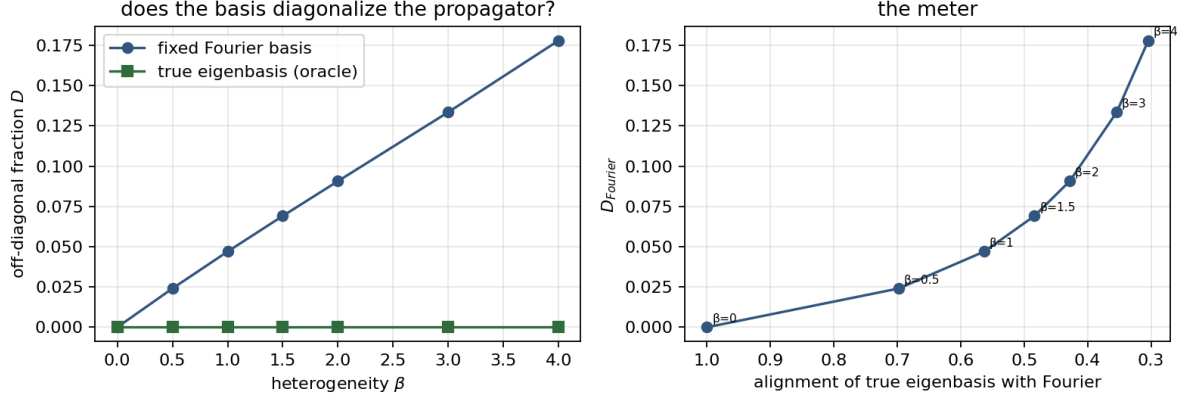


Figure 6: Heterogeneity meter. Left: off-diagonal fraction of the propagator in the fixed Fourier basis (blue) and in the true eigenbasis (green) as the medium becomes heterogeneous. Right: the same Fourier curve as a meter, against the alignment between the true eigenbasis and Fourier.

Why RK4 with dealiasing and a long horizon. Burgers is integrated with fourth-order Runge–Kutta at $dt = 2 \cdot 10^{-3}$, spatial derivatives spectral, and the quadratic term dealiased by the 2/3 rule so that the product uu_x does not fold high wavenumbers back into the resolved band. Snapshots are taken every 85 steps ($\Delta = 0.17$): a horizon long enough for the nonlinearity to act on a single step, which is what makes the DMD residual a usable meter — with a short horizon every one-step map is nearly linear and the gate never has a reason to open (Section 16). Initial conditions are smooth random superpositions of the first four harmonics at unit amplitude; 60 training and 20 test trajectories of 32 snapshots per ε , plus 20 test trajectories at amplitude 1.6 for the out-of-distribution test.

8 A heterogeneity meter

We sweep β and evaluate Equation (1) (the off-diagonality $\mathcal{D}$) for the fixed real Fourier basis and for the true eigenbasis Φ of L — an oracle, removed in the next section. At $\beta = 0$ Fourier diagonalizes P exactly ($\mathcal{D} = 0.000$). As heterogeneity grows, $\mathcal{D}_{\text{Fourier}}$ rises monotonically to 0.178 at $\beta = 4$, while $\mathcal{D}_{\Phi}$ stays at machine precision at every β ; the alignment between Φ and Fourier — the mean energy of an eigenvector in its best single frequency — falls from 1.00 to 0.30 (Figure 6). The fixed basis is optimal exactly where the medium is homogeneous, and degrades in proportion to how far the physics has moved away from it. The failure of the fixed basis is the measurement.

9 Discovering the basis without an oracle

The oracle is the weak point of the previous section. Learning a free unitary is the classification failure mode of Section 6 — under-determined and drifting — so instead we search a *structured* family: Sturm–Liouville operators $L(\theta) = -G^T \text{diag}(a_\theta)G$ with $a_\theta = 1 +$ a truncated Fourier series of M harmonics ($2M$ parameters; $M = 3$ or 5). The form is the one of Section 7: G takes a derivative, $\text{diag}(a_\theta)$ multiplies it by the coefficient, $-G^T$ takes the (negative adjoint) derivative again — together the discrete $\partial_x(a \partial_x \cdot)$. Parametrizing a_θ by a few low harmonics keeps it smooth and positive, keeps $L(\theta)$ symmetric for every θ (so its eigenbasis is always orthonormal and the off-diagonality $\mathcal{D}$ always applies), and makes $\theta = 0$ exactly the constant-coefficient operator whose eigenbasis is Fourier. Each candidate θ thus names a physical medium, and its eigenbasis is the basis that medium would prefer. The objective is only Equation (1) (the off-diagonality $\mathcal{D}$) on the true propagator: no labels, no trajectories, no oracle, started from

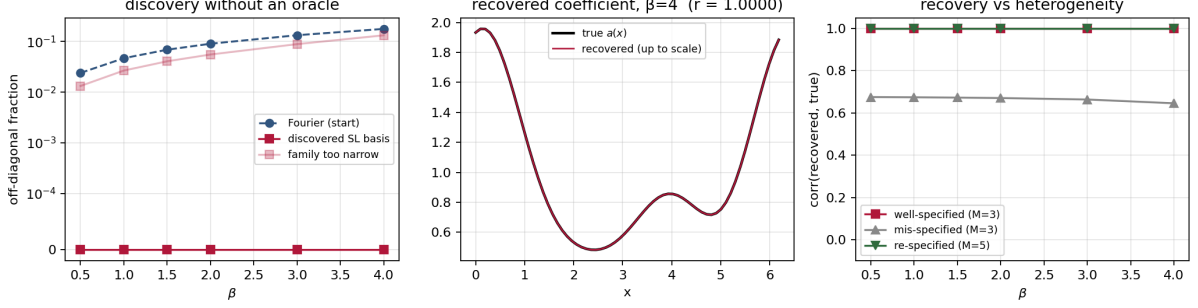


Figure 7: Discovering the basis. Left: the discovered Sturm–Liouville basis diagonalizes as well as the oracle, with no oracle; the pale curve is the same search with a family too narrow to contain the truth. Centre: the recovered coefficient over the true one. Right: recovery against heterogeneity for well-specified, mis-specified and re-specified families.

Fourier ($\theta = 0$), Powell’s method with three restarts. The family is well motivated rather than arbitrary: the classical orthogonal bases (Fourier, Hermite, Legendre, Chebyshev) are eigenbases of Sturm–Liouville operators, so searching over them is searching over second-order differential operators. A degeneracy makes the operator, not the basis, the right thing to parametrize: at a fixed rank K all polynomial families of degree $< K$ span the same subspace.

Result. Off-diagonality falls from 0.024–0.178 (Fourier) to 0.000000 at every β , and the recovered coefficient matches the truth at correlation 1.0000 (Figure 7). Eigenvectors are invariant under $a \rightarrow ca$, $c > 0$, so a is identifiable only up to a positive scale and recovery is scored after standardizing both profiles. The discovered basis is not merely a basis that works — it is the physics, readable as a coefficient field.

The meter detects its own mis-specification. When the true $a(x)$ contains a harmonic outside the search family ($M = 3$ while the truth has a fifth harmonic), recovery degrades: off-diagonality falls only $0.190 \rightarrow 0.133$ at $\beta = 4$ and the recovered profile correlates 0.65–0.68 with the truth. Widening the family to $M = 5$ restores 0.000000 and 1.0000. The failure mode is mis-specification, not the method — and the objective value announces it by plateauing well above zero. This is the precise counterweight to Section 11: structure is the lever, but wrong structure is worse than useless.

10 A nonlinearity meter, and a gate that fires

We now move to Burgers and learn a one-step map over the horizon Δ . The linear reference is the closed-form DMD propagator A (Section 3); its held-out residual is exactly what a Laplace-linear front-end cannot capture. On top we place the gated branch of Figure 3, $\hat{u}_1 = Au_0 + g \text{NL}(u_0)$, with A frozen and only the branch trained on the residual, L_1 weight 10^{-3} on g , weight decay $3 \cdot 10^{-5}$, 1500 full-batch epochs. The branch here is a black box (a two-layer perceptron); its gate is initialized open — starting it at zero makes $g \cdot \text{NL}$ a dead saddle in which neither factor receives gradient.

Two things track ε monotonically (Figure 8). The DMD residual rises $0 \rightarrow 0.173$: a clean nonlinearity meter. And the branch fires with a rectified, thresholded law — firing amplitude at or below 0.002 up to $\varepsilon = 0.2$, where it also captures nothing out of sample, then 0.013 at $\varepsilon = 0.4$, 0.052 at $\varepsilon = 1$ and 0.130 at $\varepsilon = 2$, rising roughly in proportion to the residual. That is the shape of Equation (2) (the gate law) without its saturation: shut until there is something to pay for, then proportional to it. When it fires it captures out of sample, removing 32–55% of the DMD residual for $\varepsilon \geq 0.7$.

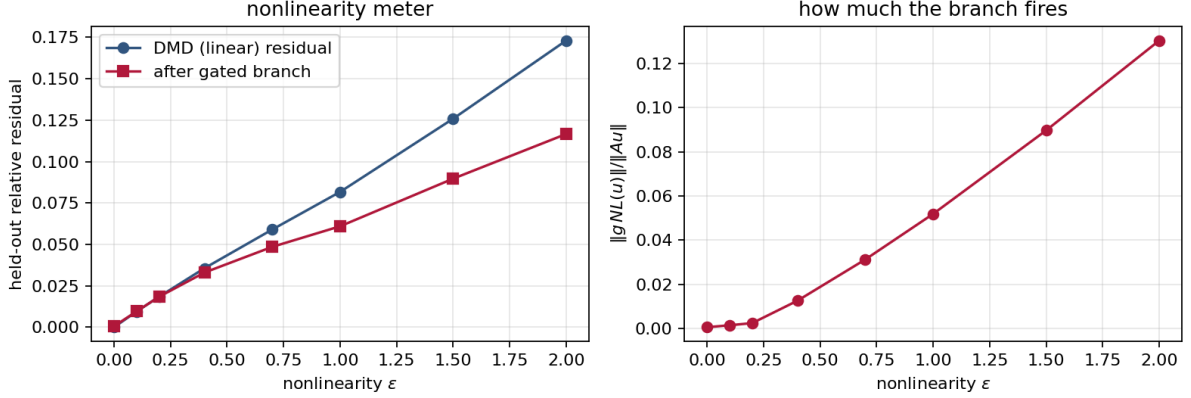


Figure 8: Nonlinearity meter and gate. Left: DMD (linear) held-out residual against the nonlinearity knob, and the residual after the gated learned branch. Right: how much the branch fires — shut below threshold, then opening in proportion.

nonlinear branch	mean held-out capture ($\varepsilon \geq 0.7$)
one-step multi-layer perceptron (no depth)	45.1%
residual integrator, 4 stages (neural-ODE-like)	68.8%
residual integrator, 8 stages	63.5%
structure-matched quadratic $[u, u_x, u u_x]$, one step	90.3%
the same term, 4 sub-steps (structured integrator)	98.9%

Table 3: Depth versus structure. Adding depth as a learned integrator helps; matching the form of the nonlinearity helps more, with a linear map and no hidden layer; sub-stepping that term closes the gap.

11 Depth is not the lever; structure is

The black-box branch above leaves a good part of the DMD residual on the table. Since uu_x is a low-order polynomial nonlinearity, the ceiling is one of generalization, not capacity. We therefore compare five branch designs on the identical frozen-DMD residual, same budget, scored by held-out capture (Table 3, Figure 9). All of them play the role of $NL(u_0)$ in $\hat{u}_1 = Au_0 + gNL(u_0)$; they differ only in what they are allowed to compute from u_0 :

- *one-step perceptron*: $u_0 \mapsto W_2 \tanh(W_1 u_0)$, a two-layer network with 64 hidden units — the generic black box, with no notion of time inside it;
- *residual integrator, k stages* ($k = 4, 8$): k small networks $f_1, \dots, f_k$ applied in sequence as $v \leftarrow v + \frac{1}{k} f_i(v)$, returning $v_k - u_0$ — the neural-ODE way of adding depth, each stage a small step along a learned vector field;
- *structure-matched quadratic, one step*: a single *linear* map W on the three features $u_0, \partial_x u_0$ and $u_0 \partial_x u_0$ (derivatives spectral), with no hidden layer; the only nonlinearity it can express is the product uu_x that Burgers actually contains;
- *structured integrator, 4 sub-steps*: the same single W applied four times in sequence, $v \leftarrow v + \frac{1}{4} W[v, v_x, v v_x]$, and returning $v_4 - u_0$. This is structure plus depth with no additional parameters: the previous term integrated in four Euler-like sub-steps instead of being applied once.

Depth buys something — the 4-stage integrator improves on the perceptron, and 8 stages under-trains at the same budget — but the structure-matched branch, which has no hidden layer at all, captures more, and it does so already at $\varepsilon = 0.1$ (81.8% for the structured integrator) where every black box is still shut (-2.2%). The winning branch is not just better: it is the

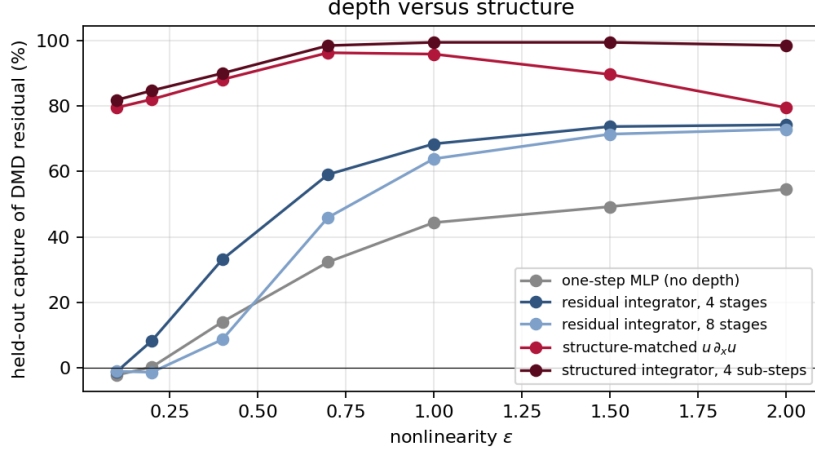


Figure 9: Held-out capture of the DMD residual for five branch designs across the nonlinearity sweep.

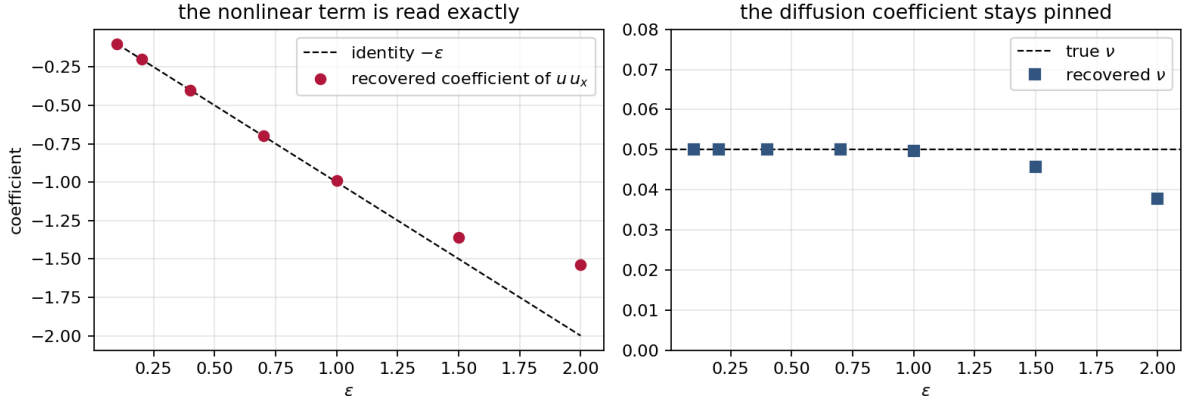


Figure 10: Reading the operator. Left: recovered coefficient of uu_x against the ground truth $-\varepsilon$. Right: the linear (diffusion) coefficient stays pinned at ν .

feature that sparse regression selects (next section). Capturing what a linear front-end misses and writing down the governing term are therefore the same operation, which is what makes the rest of Part II possible. Whether that capture is worth anything beyond one step — and whether one explicit step of the right term is enough — is the question of Section 13.

12 Reading the operator

We estimate u_t numerically (a forward difference of consecutive fine-step snapshots, not the analytic right-hand side), build a library of candidate terms $[u, u_x, u_{xx}, u_{xxx}, u^2, uu_x, uu_{xx}, (u_x)^2]$ with spectral derivatives, and run sequential thresholded least squares [5, 6], implemented directly.

The recovery is exact in the accessible regime (Figure 10): the diffusion coefficient is pinned at $0.050 = \nu$, the coefficient of uu_x lies on the identity line $-\varepsilon$ (measured $-0.200, -0.399, -0.699, -0.988$ for $\varepsilon = 0.2, 0.4, 0.7, 1.0$), and every non-physical library term is driven to zero by the sparsity. Honest limit: for $\varepsilon \geq 1.5$ the nonlinear coefficient is under-estimated (-1.537 against -2.0) and ν drifts to 0.038 . This is a resolution effect — strong shocks are under-resolved at $N = 64$ and the numerical u_t degrades where gradients steepen — not a failure of identification.

model	rollout error, in-distribution	rollout error, amplitude 1.6
DMD linear only	0.697	0.860
DMD + black-box branch	0.978	2.328
DMD + discovered uu_x , one explicit step	0.063	diverged
DMD + discovered uu_x , 4 sub-steps	0.039	0.102

Table 4: Relative L_2 error at rollout step 30, seed 0, one-step training only. The structured integrator is $19\times$ better than the linear model in horizon and $9\times$ at unseen amplitude; the black box is worse than the linear model on both.

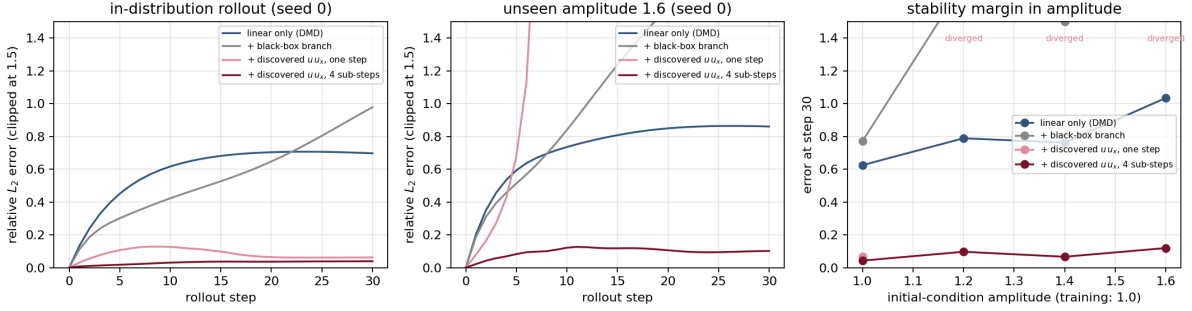


Figure 11: Extrapolation. Rollout error against step in distribution (left) and at amplitude 1.6 (centre); error at step 30 against initial-condition amplitude, fresh initial conditions (right). Curves are clipped at 1.5; "diverged" marks errors that left the plot.

13 What the reading is worth: extrapolation

Recovering an equation is only valuable if it buys something a black box cannot give. We test the sharpest version: one-step models sharing the same frozen DMD head, differing only in the nonlinear branch — (a) none, (b) the black-box perceptron, (c) the discovered uu_x term in one explicit step, (d) the same term sub-stepped four times — all trained only on one-step pairs at $\varepsilon = 1.0$.

What is measured. A *rollout* feeds the model its own output: from a true initial state u_0 the model predicts $\hat{u}_1$, then $\hat{u}_2 = \text{model}(\hat{u}_1)$, and so on for 30 steps ($30\Delta = 5.1$ time units), with no correction from the data in between. The *rollout error* at step s is the relative L_2 error $\|\hat{u}_s - u_s\|/\|u_s\|$ against the reference RK4 trajectory, pooled over the test initial conditions; Tables 4 and 5 report it at $s = 30$, the figures against s . Every model was trained on single steps only, so 30 steps is a horizon it never saw. *In-distribution* means the 20 test initial conditions are drawn from the same distribution as training (smooth, unit amplitude, harmonics 1–4) but were never used in training. *Amplitude 1.6* means the same kind of initial condition scaled to $1.6\times$ the training amplitude, i.e. a regime the models never saw at all, in which Burgers' quadratic term is $1.6^2 \approx 2.6\times$ stronger; a further sweep uses fresh initial conditions at amplitudes 1.0, 1.2, 1.4 and 1.6 (Table 4, Figure 11).

The question the table answers is therefore: of four models that are comparably good at what they were trained on (one step), which keeps being good at what they were not trained on (many steps, larger waves)? Three things are visible. First, the black-box branch captured 44.4% of the one-step residual at $\varepsilon = 1$ and that capture does not compound: rolled out, it is *worse* than the linear model it was meant to correct, in distribution and more so at larger amplitude, and it creates energy on 14% of its in-distribution steps (33% out of distribution), which viscous Burgers never does. It learned a one-step correction on the training distribution, not an operator. Second, the discovered term applied in one explicit step extrapolates in horizon (error 0.063 at step 30) but has no stability margin in amplitude: at every amplitude above training it diverges

model	in-distribution	unseen amplitude
black-box branch, one-step trained	1.152 ± 0.249	2.907 ± 0.409
black-box branch, + rollout training	0.531 ± 0.063	1.508 ± 0.124
structured integrator, one-step trained	0.037 ± 0.002	0.096 ± 0.009
structured integrator, + rollout training	0.024 ± 0.009	0.047 ± 0.004

Table 5: Rollout error at step 30, three seeds (network initialization only). Rollout training improves the structured integrator $1.6\times$ in distribution and $2.1\times$ out of distribution, and the black box $2.2\times$ and $1.9\times$ — which leaves it worse than the linear model at unseen amplitude.

(right panel). This is the behaviour of an explicit integrator pushed past its stability limit — the term is right, the step is too long for it — and it is a negative result we did not anticipate. Third, sub-stepping the same term restores the margin: the structured integrator stays at 0.102 at $1.6\times$ amplitude and below 0.120 across the sweep, while the linear model reaches 1.034, and it never creates energy. The structured term transfers because uu_x is amplitude-covariant — under $u \rightarrow cu$ it scales as c^2 exactly as the physics does — so the physically correct term carries to a regime it never saw; but only a model that integrates that term, rather than applying it once, can use the covariance.

14 Hardening: rollout training multiplies structure

Section 13 trained every model on single steps and only *tested* it on rollouts. The obvious objection is that a model trained on rollouts might do better — that the black box failed because it was never asked to be stable, not because it lacks structure. This section tests that objection. *Rollout training* means the loss is computed on the model’s own multi-step predictions: from each training state we unroll 5 steps, $\hat{u}_1, \dots, \hat{u}_5$, compare each to the true trajectory, and back-propagate through the whole chain, so the model is penalized for errors that compound. We apply it as a *fine-tuning* stage on top of the already converged one-step model (300 epochs at a lower learning rate, gradients clipped to norm 1), because training on rollouts from a cold start diverged (Section 16). Everything else is as in Section 13: same frozen DMD head, same test and out-of-distribution initial conditions, same 30-step rollout error. The comparison is one-step-trained versus rollout-trained, for the black box and for the structured integrator, each over three seeds, mean $\pm$ standard deviation (Table 5, Figure 12). The one-step structured surrogate is included for completeness: rollout training improves it in distribution ($0.064 \pm 0.001 \rightarrow 0.033 \pm 0.001$) and does not rescue it out of distribution, where it still diverges.

What the seeds vary. The three seeds change the initialization of the nonlinear branch and the order of stochastic operations in training; they do *not* resample the data. The training trajectories, the test initial conditions and the out-of-distribution initial conditions are fixed across seeds. The error bars therefore measure training variance, not data variance — the honest reading is "how reproducible is this model on this data", and the limitation is stated in Section 16.

Rollout training is not a substitute for structure; it is a multiplier on it. It improves both models by a similar factor, and a similar factor applied to 2.907 ± 0.409 and to 0.096 ± 0.009 leaves one model unusable and the other at 0.047 ± 0.004 . The physical check — the fraction of rollout steps on which the predicted energy grows, which for viscous Burgers must be zero — discriminates here: the structured integrator never creates energy (0% of steps in and out of distribution, before and after rollout training), the linear model never does either (0%), the black box does on 14% of its in-distribution steps after one-step training and still on 11% of its out-of-distribution steps after rollout training, and the one-step structured surrogate does on 66% of its out-of-distribution steps as it diverges. We report it as a check, not as a metric optimized for: nothing in training penalized energy growth.

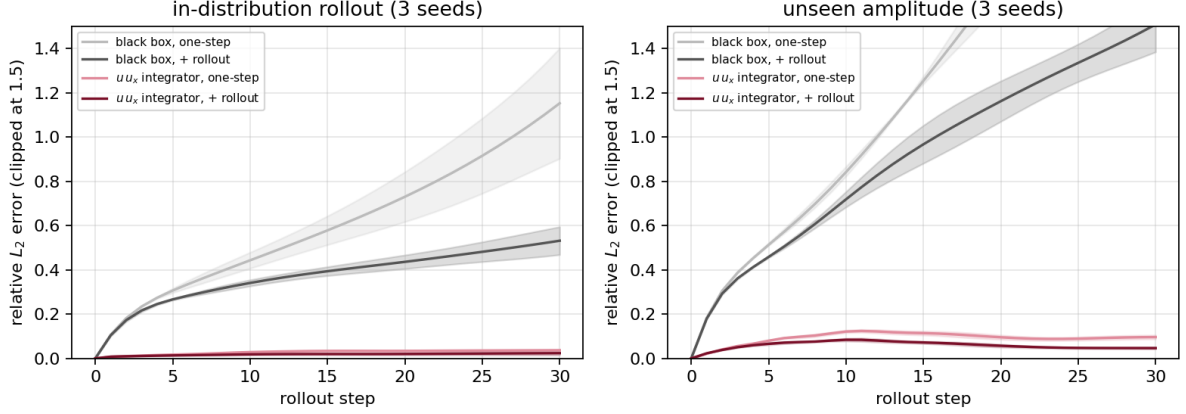


Figure 12: Hardening, three seeds (shaded: ± 1 s.d.; curves clipped at 1.5). Rollout training helps both models; only the one with the right structure ends up usable.

15 Related work

Learning solution operators for PDEs is an active field; Fourier neural operators [1] are, in the language of this paper, the fixed-spectral branch promoted to an operator. Koopman theory and DMD [2, 3] are the learned-linear branch, and the reduced-order-modelling literature has long used them for prediction; Prony’s method [4] is their ancestor for sums of exponentials. Sparse identification of nonlinear dynamics [5] and PDE-FIND [6] provide the reading step of Section 12; we use them unmodified. Recovering a coefficient field from spectral data is the classical inverse-conductivity problem [7], a mature area; Section 9 is a small, well-specified parametric instance of it and claims no advance there. On the classification side, the information bottleneck [9] is the formal statement that compressing toward a label and toward a signal are different objectives; Section 4 measures a gate against that gap rather than proposing a new bound. It has been proved that invariant networks recover the Fourier transform of the underlying group and that the group is readable from the weights [8]; that result is stronger than anything we show about why classical bases appear, and we position our contribution as complementary and empirical. What we believe is not covered by the above is the packaging: a small set of calibrated meters with a shared shape — fixed structure plus measured failure — read through the *same* gate in two objectives, together with the controls that keep them honest.

16 Limitations and negative results

We state these explicitly, in the order they cost us time.

Two settings are two. The gate law has the same functional form in classification and in physics. That is a coincidence of shape observed in two constructed families, not a universal law; we have not derived it, and $n = 2$ does not license the word "universal". What we claim is narrower: in both settings the L_1 -gated branch behaves as a rectified meter of what the branch adds, and that reading survived the controls below.

An artefact we had to remove. The heterogeneity meter was first built from a prediction error with Fourier-smooth initial conditions. It produced a negative gap — Fourier apparently better than the eigenbasis — because those initial conditions let Fourier compress the data irrespective of the dynamics, and a full linear head made the basis irrelevant. Moving to the operator-level metric of Equation (1), the off-diagonality $\mathcal{D}$, removed the confound.

An unfair comparison we had to fix. The first gated experiment trained a coupled network by stochastic gradient descent against a closed-form least-squares DMD and unsurprisingly lost, with a gate that did not behave as a meter. Freezing the DMD operator and training only the gated branch on its residual is what turned the gate into an instrument. A related recipe detail: a gate initialized at zero never opens, because $g \cdot \text{NL}$ is then a saddle in which neither factor receives gradient; the gate must start open and be pushed shut by the penalty.

A horizon that hid the nonlinearity. With a short snapshot interval every one-step map is nearly linear, the DMD residual is tiny and the L_1 penalty crushes the gate. The horizon must be long enough for the nonlinearity to act within one step.

A training recipe that diverges. Rollout training from a cold network diverges (in an earlier run we observed exactly 1.000 relative error, i.e. a collapsed prediction). It must be applied as fine-tuning on a converged one-step model, with gradient clipping.

The right term in one step is not enough. We expected the one-step structured surrogate to extrapolate in amplitude because its term is amplitude-covariant. It does not: it diverges at every amplitude above training, like an explicit integrator past its stability limit. Sub-stepping the same term fixes it with no new parameters, but this was found, not designed, and the stability margin of the structured integrator has been mapped only up to $1.6\times$ the training amplitude.

Seeds and data. Out-of-distribution rollout numbers vary across seeds for the black box; single-seed out-of-distribution claims should not be trusted, which is why Section 14 reports error bars and Section 13 should be read through them. The seeds vary the network, not the data; a data-resampled study would be the next step.

The information estimate is a bound. $I(T; Y)$ is estimated by $H(Y)$ minus a held-out cross-entropy, a lower bound that depends on the head; we use the same small head for both branches so that the *gap* is comparable, and we clip negative estimates to zero. A kNN or MINE estimator would be a useful cross-check.

Scope. All physics experiments are one-dimensional, at $N = 64$, with a single nonlinear term whose form we ourselves planted. Recovery correlations of 1.0000 indicate a well-specified problem, not a difficult one: they validate the instrument, they do not demonstrate discovery on data whose answer is unknown. Strong shocks are under-resolved. The linear head is a fitted DMD operator, not the exact diffusion operator, which caps how far even the structured model transfers in amplitude. Structure preservation is verified a posteriori (energy monitoring) rather than imposed by construction; a symplectic or conservative parametrization is the natural next hardening step. The classification family is synthetic by design, so that the gap is controlled; the four benchmark tasks are the only uncontrolled points.

17 Conclusion

A gate that decides how much of a learned branch to use is worth exactly what the branch adds, and the two settings studied here agree on the shape of that price: nothing until there is something to pay for, then proportional. In classification the currency is information about the label, and the learned part cannot be read back because many bases pay the same. Inverting the objective to physical modelling makes the learned branch legible: the failure of a fixed spectral basis is a heterogeneity meter; the residual of a linear propagator is a nonlinearity meter; the branch that captures that residual best is not the deepest but the one with the right structure;

that structure is the term sparse regression names; and putting the named term into the model — integrated, not applied once — is what buys extrapolation in horizon, in amplitude, and under rollout training, where structure is multiplied rather than replaced. The practical reading for a modeller is short: measure first whether your fixed front-end is failing and in which way, prefer a structured learned family over a free one, and treat the objective value itself as a mis-specification alarm.

18 Reproducibility

NumPy/SciPy/PyTorch with fixed seeds, CPU only. Every experiment is a short deterministic script that regenerates the figures of the same name and writes its numbers to a JSON file in `results/`; every number in this paper is read from those files by `tools/make_numbers.py`, which writes the macros used in the text, so a re-run that changes a number changes the paper. The classification scripts import the unmodified S1/S2/S3 code of [10], vendored under `classification/flob_lib/`. Trajectory generation for the Burgers sweep is cached in `results/cache/` and recomputed deterministically when absent.

section	script(s)
3 DMD as Laplace	<code>physics/dmd_laplace_check.py</code>
4 gate law in bits	<code>classification/gap_law.py</code>
5 rate-relevance	<code>classification/rate_relevance.py</code>
6 reading the basis	<code>classification/symbolic_U.py</code>
8, 9 heterogeneity, discovery	<code>physics/phys_wp1.py</code>
10 nonlinearity meter	<code>physics/phys_wp2.py</code>
11 depth versus structure	<code>physics/depth_test.py</code>
12 reading the operator	<code>physics/phys_wp3_sindy.py</code>
13, 14 extrapolation, hardening	<code>physics/phys_wp4_rollout.py</code>

Table 6: Section-to-script map. Shared numerics: `physics/common.py`, `physics/data.py`, `physics/models.py`.

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